

Toward a uniqueness theorem for the container derivative the solid infinitesimal parts of polynomial functors

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3 October 2026

Abstract

A companion note showed that the Abbott–Altenkirch–Ghani–McBride one-hole derivative ∂ is the differential combinator of a Cartesian differential category of polynomial functors, realised by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$. That settles *existence*. This note asks for *uniqueness*: is ∂ the only nontrivial first-order tangent structure a polynomial functor can carry? We prove the decisive rigidity input. An infinitesimal object $D = y \oplus M$ in the substitution monoidal category $(\mathbf{Poly}, \triangleleft, y)$ has, for a tangent bundle $T = (-) \triangleleft D$, a *solid* infinitesimal part when its multiplication $M \triangleleft M \rightarrow M$ is invertible; we show that over the rig $\mathbb{N}$ the linear containers are automatically free, so solidity reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1. Hence there are *exactly two* finite-support solid infinitesimal parts, $M = 0$ and $M = y$, corresponding to the trivial structure and to ∂ . The finiteness is sharp: $M = \mathbb{N} \cdot y$ is a third solid part, since $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$. This is the polynomial-functor analogue of Lanfranchi–Lemay’s rigidity for affine schemes, and cleaner — no principal-ideal hypothesis is needed. Uniqueness of the tangent structure itself follows once one knows that every representable first-order tangent structure has a solid infinitesimal part; we verify this converse on the two primitives and against the Whitney sums, and state the general case as the one open problem that stands between this classification and a full uniqueness theorem.

1 Introduction

From existence to uniqueness. In a companion note [8] we showed that the category $\mathbf{SPoly}$ of finitary polynomial functors is a Cartesian differential category whose differential combinator is the Abbott–Altenkirch–Ghani–McBride one-hole-context derivative ∂ [1], and that the attached tangent bundle is computed by the oldest trick in differential algebra: substitute a dual number and read off the ε -coefficient,

$$T(p) = p(y + \varepsilon v) \bmod \varepsilon^2 = p(y) + \varepsilon v \cdot \partial p.$$

That note answers an *existence* question — there is a tangent structure on polynomial functors whose vertical part is ∂ . The natural next question is *uniqueness*. The dual-number structure is one tangent structure; the identity functor $T = \text{id}$ is another (the trivial one). Are there others? Could a polynomial functor carry an exotic first-order tangent bundle that is not the derivative and not trivial?

The template. For commutative algebra this question has a sharp answer. Lanfranchi and Lemay [7] classify the *representable* tangent structures on the opposite of the category of modules (equivalently, affine schemes): a representable tangent structure is a tensoring $T = D \otimes (-)$ by an

infinitesimal object $D = R \oplus M$, and when the base ring R is a principal ideal domain there are *exactly two* — the trivial one ($M = 0$) and the Kähler/dual-number one ($M = R$). The mechanism is a rank computation: the infinitesimal part M of a tangent structure must be *solid*, $M \otimes_R M \cong M$; over a PID a solid finitely generated module is free of rank n with $R^{n^2} \cong R^n$, forcing $n^2 = n$, hence $n \in \{0, 1\}$ (their Theorem 4.17, Corollary 4.18).

The port, and what is genuinely proved. This note ports that mechanism to polynomial functors, and the port is in one respect *cleaner* than the original. The substitution monoidal category $(\text{Poly}, \triangleleft, y)$ plays the role of $(\text{Mod}_R, \otimes_R, R)$; an infinitesimal object is a polynomial $D = y \oplus M$ with an augmentation, and the representable tangent bundle is $T = (-) \triangleleft D$. The infinitesimal part M is forced to be a *linear* container $M = S \cdot y$ by the additive-bundle axiom, and here is the simplification: over the rig $\mathbb{N}$ a linear container *is* a free module, with no principal-ideal hypothesis to invoke, because the linear polynomial functors are by definition the free $\mathbb{N}$ -modules. Solidity then reads $|S|^2 = |S|$ in the cardinal $|S| = \text{rank } M$, whose only finite solutions are 0 and 1. We prove:

Theorem 1 (Classification of finite solid infinitesimal parts; §4). *A finite-support solid infinitesimal part of $(\text{Poly}, \triangleleft, y)$ has rank in $\{0, 1\}$. There are exactly two: $M = 0$, giving the trivial structure $T = \text{id}$, and $M = y$, giving the AAGM dual-number structure $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ with combinator ∂ .*

This is the airtight core, and its arithmetic is machine-checked (§4, a Mathlib-free Lean 4 oracle with no classical axioms).

The honest gap. Theorem 1 classifies the solid *infinitesimal parts*. To conclude that there are exactly two tangent *structures* one needs the correspondence

$$(\star) \quad \text{representable first-order tangent structure } T = (-) \triangleleft D \iff \text{solid infinitesimal part } M = \ker(D \rightarrow y).$$

The forward direction (a solid part yields a tangent structure) we verify on the two primitives, which is all Theorem 1 leaves to check. The converse (a tangent structure forces its infinitesimal part solid) is the content of Lanfranchi–Lemay’s vertical-lift universality (their Theorem 4.8), proved over a commutative ring; since $\mathbb{N}$ is a rig, their argument does *not* transfer verbatim, and we do not reprove it here. We therefore state the resulting uniqueness as a theorem *conditional on* $(\star_c)$ (the converse), and flag $(\star_c)$ as the one open problem (§6, Problem 10). We support it with two pieces of evidence: the forward correspondence on primitives, and the observation that the only other finite candidates — the higher square-zero objects $D = y + \sum_{i=1}^r \varepsilon_i v_i$, $r \geq 2$ — are the Whitney sums $T_r = T \times_X \cdots \times_X T$, which are auxiliary bundles in the tangent axioms, not standalone structures, and which are indeed non-solid ($\text{rank} = r^2 \neq r$). So solidity coincides with being a primitive tangent structure, exactly as the correspondence predicts.

The finiteness is sharp. Unqualified uniqueness over all of Poly is false, and we exhibit the obstruction explicitly: $M = \mathbb{N} \cdot y$ is solid, because $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y$ by the Cantor pairing (§5, Proposition 9). This is the precise analogue of Lanfranchi–Lemay’s extra structures when finite generation is dropped (their Examples 4.20–4.21), and it tells us the right hypothesis is a *finite-dimensional* infinitesimal part — finitely many tangent directions — here taken as plain finiteness of the shape set.

Contribution. We do not claim a full uniqueness theorem; we claim its rigidity half. The contributions are: the classification of finite solid infinitesimal parts (Theorem 1), proved over a rig with no PID hypothesis and Lean-certified in its arithmetic; the single technical observation that makes the port faithful — that on linear containers the *bundle* tensor $\triangleleft$ and the *module* (Dirichlet) tensor coincide (Lemma 4), so solidity is computed in the bundle’s own tensor; the sharpness of the finiteness hypothesis (Proposition 9); and the precise reduction of full uniqueness to the one ported converse ($\star_c$). The result is timely for the tangent-category and **Poly** communities: “we have classified the solid infinitesimal parts of finite-support polynomial functors, and uniqueness of the container-derivative tangent structure reduces to vertical-lift universality” is a clean statement of where the problem stands.

Outline. §2 recalls containers, the substitution monoidal structure, infinitesimal objects and the representable tangent bundle. §3 proves the bundle/module tensor coincidence. §4 proves the rank rigidity (Theorem 1) and records the Lean certification. §5 proves the infinite-support obstruction. §6 discusses the correspondence ($\star$), states the conditional uniqueness theorem and the open converse. §7 compares with Lanfranchi–Lemay and states the significance. §8 concludes.

2 Preliminaries

We recall only what the rigidity argument uses; for the Cartesian differential and tangent structure on **SPoly** we refer to the companion note [8].

Containers and the substitution monoidal category. A *container in n variables* is a set S of *shapes* together with, for each $s \in S$ and each $i \in \{1, \dots, n\}$, a position set P_s^i ; its extension is the polynomial functor $\llbracket S \triangleleft P \rrbracket(\vec{X}) = \sum_{s \in S} \prod_{i=1}^n X_i^{P_s^i}$. We work with finitary containers, so only the cardinalities $|P_s^i|$ matter up to isomorphism [1]. Composition of containers is substitution of extensions, written $G \triangleleft F$, with unit the identity container y ; this makes $(\mathbf{Poly}, \triangleleft, y)$ a (strict, once we pass to isomorphism classes) monoidal category, and the comonoids in it are exactly the small categories [2, 6]. A container is *linear* if it is of the form $S \cdot y$ for a set S , i.e. its extension is $\vec{X} \mapsto S \times X$ in one variable; *finite-support* if its shape set is finite.

Infinitesimal objects and representable tangent bundles. We use the representable presentation of tangent structure of [7], transported along the dictionary $(\mathbf{Mod}_R, \otimes_R, R) \rightsquigarrow (\mathbf{Poly}, \triangleleft, y)$.

Definition 2 (Infinitesimal object). An *infinitesimal object* in $(\mathbf{Poly}, \triangleleft, y)$ is a polynomial D equipped with an augmentation: morphisms $z: y \rightarrow D$ (zero) and $p: D \rightarrow y$ (projection) with $p \circ z = \text{id}_y$. The splitting $D = y \oplus M$ defines the *infinitesimal part* $M = \ker(p)$. Right- $\triangleleft$ -tensoring by D is the endofunctor $T := (-) \triangleleft D$, with tangent projection $A \triangleleft p: A \triangleleft D \rightarrow A$ and zero section $A \triangleleft z: A \rightarrow A \triangleleft D$.

Definition 3 (Representable first-order tangent structure). A *representable first-order tangent structure* is a tangent structure (in the sense of Cockett–Cruttwell [5], first order) whose bundle functor is $T = (-) \triangleleft D$ for an infinitesimal object $D = y \oplus M$. Its infinitesimal part M is called *solid* if the tangentoid multiplication $\mu: M \triangleleft M \rightarrow M$ is an isomorphism.

The dual-number point $D = y + \varepsilon v$ (read as the infinitesimal object of the base change $\mathbb{N} \rightarrow \mathbb{N}[\varepsilon]/\varepsilon^2$) is such an object: the augmentation is $\varepsilon \mapsto 0$, the zero is $y \hookrightarrow D$, and $(-) \triangleleft D$ is exactly the dual-number substitution T of [8]. Its infinitesimal part is the linear container $M = \langle \varepsilon v \rangle = 1 \cdot y$,

of rank 1; the trivial structure is $D = y$, $M = 0$, $T = \text{id}$. That $T = (-) \triangleleft D$ is a genuine tangent structure — not merely an endofunctor — is the content of [8]; we take representability of ∂ as established there and ask only whether other infinitesimal objects give rise to further structures.

Why the infinitesimal part is linear. For a tangent structure the projection $p: T \Rightarrow \text{id}$ is an *additive bundle*: each fibre carries a commutative-monoid structure and the bundle addition is linear. For $T = (-) \triangleleft D$ this forces the infinitesimal part M to be a *linear* container $M = S \cdot y$. A non-linear M — for instance εy^2 , whose ε -coefficient is $y^2 p'$ rather than a directional $v \cdot p'$ — has no vector-space fibre and fails the additive-bundle axiom. So linearity of M is *forced*, not assumed, and it is this linearity (not any projective-module theory) that supplies freeness in §4.

3 The bundle and module tensors coincide on the linear layer

The port of the Lanfranchi–Lemay argument has one notorious hazard: the tangent *bundle* is built from substitution $\triangleleft$, whereas the *solid* condition $M \otimes M \cong M$ lives in the “module layer” where rank multiplies, and these look like two different monoidal products. One might fear a mismatch between the tensor the bundle uses and the tensor in which solidity is measured. The following observation dissolves the fear.

Lemma 4 (Tensor coincidence on the linear layer). *For linear containers, substitution and the Dirichlet product agree:*

$$(S \cdot y) \triangleleft (T \cdot y) = (S \times T) \cdot y = (S \cdot y) \otimes_{\text{Dir}} (T \cdot y).$$

Proof. The extension of $S \cdot y$ is $X \mapsto S \times X$. Composing, $(S \cdot y) \triangleleft (T \cdot y): X \mapsto S \times (T \times X) = (S \times T) \times X = (S \times T) \cdot y$. The Dirichlet product of $\sum_s y^1$ and $\sum_t y^1$ is $\sum_{(s,t)} y^{1 \times 1} = (S \times T) \cdot y$. Both equal $(S \times T) \cdot y$. $\square$

Remark 5. The two tensors agree on monomials ($y^a \triangleleft y^b = y^{ab} = y^a \otimes_{\text{Dir}} y^b$), and in particular on the linear layer, which is all the rigidity needs: the infinitesimal part is linear (§2), and $M \triangleleft M$ is computed there. On polynomials with several monomials the two diverge — already $(y + 1) \triangleleft (y + 1) = y + 2$ while $(y + 1) \otimes_{\text{Dir}} (y + 1) = y + 3$ — so Lemma 4 is genuinely a statement about the linear layer, not a global identity. Neither tensor is the “doubling” Dirichlet action $(2y) \otimes_{\text{Dir}} (-)$, which sends $p \mapsto 2p$ and has nothing to do with the derivative.

Consequence. On the linear layer the *bundle* tensor $\triangleleft$ already multiplies rank, so the solid condition $M \triangleleft M \cong M$ reads $|S|^2 = |S|$ in the very tensor the bundle uses. There is no mismatch; this single coincidence is what makes the port faithful.

4 Rank rigidity

We now classify the finite solid infinitesimal parts. Two structural facts about linear containers, then the rank computation.

Lemma 6 (Freeness is automatic over $\mathbb{N}$). *Every linear container $S \cdot y$ is the free $\mathbb{N}$ -module on the set S ; two linear containers are isomorphic iff their shape sets are equinumerous. The cardinal $|S| =: \text{rank}(M)$ is a complete isomorphism invariant.*

Proof. Under coproduct of containers and the $\mathbb{N}$ -action $n \cdot (S \cdot y) = (n \times S) \cdot y$, the linear containers form the free- $\mathbb{N}$ -module functor applied to sets: $S \cdot y$ is the $|S|$ -fold copower of y . A container isomorphism $S \cdot y \cong T \cdot y$ is, since morphisms between linear containers are just functions on shapes (positions being singletons), a bijection $S \cong T$. Hence rank is a complete invariant. $\square$

This is the analogue of Lanfranchi–Lemay’s “projective $\Rightarrow$ free $\Rightarrow$ rank is a complete invariant” (their Lemma 4.16, via Rotman’s structure theorem, which *requires* the base to be a PID). Here it is *automatic*: the linear polynomial functors *are* the free $\mathbb{N}$ -modules by definition, so no PID hypothesis is needed and the rigidity below holds over the rig $\mathbb{N}$ — strictly outside the Lanfranchi–Lemay hypotheses.

Lemma 7 (Solid $\Rightarrow$ idempotent rank). *If $M = S \cdot y$ is solid, i.e. $\mu: M \triangleleft M \rightarrow M$ is an isomorphism, then $|S|^2 = |S|$.*

Proof. By Lemma 4, $M \triangleleft M = (S \times S) \cdot y$, of rank $|S|^2$ by Lemma 6. An isomorphism $\mu: (S \times S) \cdot y \cong S \cdot y$ forces, by Lemma 6, a bijection $S \times S \cong S$, i.e. $|S|^2 = |S|$. $\square$

Theorem 8 (Finite rigidity). *A finite-support solid infinitesimal part has rank in $\{0, 1\}$. Hence there are exactly two: $M = 0$ (rank 0) and $M = y$ (rank 1). By §2 these are the trivial structure and the AAGM dual-number structure, with combinator ∂ .*

Proof. For a natural number $n = |S|$ we have $n^2 = n \iff n(n-1) = 0 \iff n \in \{0, 1\}$, since $\mathbb{N}$ has no zero divisors. By Lemma 7 a finite solid part has $|S|^2 = |S|$, so $|S| \in \{0, 1\}$; by Lemma 6 the rank determines M , giving $M \in \{0, y\}$. $\square$

This is the Lanfranchi–Lemay rank computation $R^{n^2} \cong R^n \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$, ported verbatim with the cardinality rank, and — per Lemma 6 — without their PID hypothesis.

Machine certification of the arithmetic. The finite arithmetic of Theorem 8 is certified in a Mathlib-free Lean 4 development with no classical axioms. The oracle proves: $n^2 = n \iff n \in \{0, 1\}$ on $\mathbb{N}$ (in general, *omega-free*); that the solid ranks in $0 \leq n \leq 12$ are exactly $\{0, 1\}$ (by *decide*); that these two ranks map through the structure assignment to exactly $\{\text{id}, \partial\}$, with the dual-number object of arity 2; that every rank $r \geq 2$ (the Whitney sums of §6) is non-solid; and that the Cantor pairing is injective on a 12×12 window, witnessing $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ for §5. *#print axioms* on the packaged oracle reports only *propext* (a structural, non-classical axiom); the finite *decide* claims are individually axiom-free, with no *sorryAx*, *Quot.sound* or *Classical.choice*. The Lean certifies the *arithmetic collapse* only — not the structural correspondence of §6, which remains the open part.

5 The finiteness hypothesis is sharp

Theorem 8 is stated for finite support, and the hypothesis cannot be dropped.

Proposition 9 (Infinite-support obstruction). *The countable linear container $M = \mathbb{N} \cdot y$ is solid: $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y = M$, because $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$. It is a solid infinitesimal part distinct from 0 and y . Consequently unqualified uniqueness over all of Poly is false.*

Proof. Lemma 4 gives $M \triangleleft M = (\mathbb{N} \times \mathbb{N}) \cdot y$, and the Cantor pairing supplies a bijection $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$ (injectivity checked on a 40×40 window; see the Lean oracle of §4 for a certified 12×12 window). By Lemma 6 this is an isomorphism $\mu: M \triangleleft M \cong M$, and $\text{rank } M = |\mathbb{N}| \notin \{0, 1\}$. $\square$

More generally every infinite cardinal κ satisfies $\kappa^2 = \kappa$ (under choice), so each yields a solid infinitesimal part. This is the exact analogue of Lanfranchi–Lemay’s extra structures when the finite-generation hypothesis is relaxed — their non-PID example $R = S \times S$ (Example 4.21) and non-finitely-generated $\mathbb{Z} \ltimes \mathbb{Q}$ (Example 4.20). The moral is that the honest hypothesis is a *finite-dimensional* infinitesimal part: finitely many tangent directions, matching Lanfranchi–Lemay’s finite-generation (Niefield coexponentiability) condition, here realised as plain finiteness of the shape set.

6 From solid parts to tangent structures

Theorem 8 classifies the finite solid infinitesimal *parts*: exactly $\{0, y\}$. To pass from parts to tangent *structures* one needs the correspondence

($\star$) representable first-order tangent structure $T = (-) \triangleleft D \iff$ solid infinitesimal part $M = \ker(D \rightarrow y)$, with the augmented square-zero structure.

This is Lanfranchi–Lemay’s Theorem 3.9 (tangentoid $\iff$ representable tangent structure) composed with their Theorems 4.8/4.10 (such a tangentoid’s infinitesimal part is solid). Over a commutative ring the correspondence is an equivalence; over the rig $\mathbb{N}$ it splits honestly into a forward direction we can verify and a converse we cannot yet reprove.

Forward: solid $\Rightarrow$ structure. A finite solid part yields a tangent structure in the two surviving cases. $M = 0$ gives $T = \text{id}$; $M = y$ gives the AAGM dual-number structure, which *is* a tangent structure — the full Cartesian differential / tangent verification is the content of [8]. Since Theorem 8 leaves no third finite solid part, the forward direction never has to manufacture a third structure.

Converse: structure $\Rightarrow$ solid. This is the heart of Lanfranchi–Lemay Theorem 4.8: the *universality of the vertical lift* forces the tangentoid multiplication to be invertible (solid). Their proof is module-theoretic but is packaged over a commutative ring; the step that repackages the tangentoid as a square-zero commutative algebra (their Theorem 4.10) uses ring subtraction, which $\mathbb{N}$ lacks. We therefore *do not* reprove the converse rig-internally. We state it as the open problem that stands between Theorem 8 and a full uniqueness theorem.

Problem 10 (The solidity converse). Let $T = (-) \triangleleft D$ be a representable first-order tangent structure on finite-support polynomial functors, with infinitesimal part $M = \ker(D \rightarrow y)$. Show, internally to $(\text{Poly}, \triangleleft, y)$ over the rig $\mathbb{N}$, that the universality of the vertical lift forces $\mu: M \triangleleft M \rightarrow M$ to be an isomorphism (i.e. M is solid).

We offer two pieces of evidence that Problem 10 resolves affirmatively.

Evidence 1 (the non-solid candidates are not primitive structures). The finite candidate objects not covered by Theorem 8 are the higher square-zero objects

$$D_r = y + \varepsilon_1 v_1 + \cdots + \varepsilon_r v_r, \quad \varepsilon_i \varepsilon_j = 0, \quad r \geq 2,$$

with infinitesimal part $M = \mathbb{N}^r$, so $M \triangleleft M = \mathbb{N}^{r^2}$ of rank $r^2 \neq r$: *not solid*. But these D_r are precisely the Whitney sums $T_r = T \times_X \cdots \times_X T$, which occur in the tangent axioms as *auxiliary* bundles, not as standalone tangent structures. So the non-solid finite candidates are exactly the

ones that are not primitive tangent structures: solidity coincides with indecomposability, as the correspondence predicts. (The non-solidity of every $r \geq 2$ is part of the certified Lean oracle of §4.)

Evidence 2 (the module layer is the one the converse uses). The module layer of polynomial functors is the free- $\mathbb{N}$ -module layer, carrying the same symmetric monoidal structure (Lemma 4) and augmentation that Lanfranchi–Lemay’s Theorem 4.8 uses. The place their argument invokes ring subtraction is the CALG repackaging (Theorem 4.10), not the solidity derivation (Theorem 4.8), which is module-theoretic throughout. This is why we expect a rig-internal proof to exist; constructing it is Problem 10.

Granting $(\star_c)$ — the converse — we obtain the conditional uniqueness theorem.

Theorem 11 (Conditional uniqueness). *Assume the solidity converse of Problem 10. Then among representable first-order tangent structures $T = (-) \triangleleft D$ on finite-support polynomial functors there are exactly two: the trivial structure ($D = y$, $T = \text{id}$) and the AAGM dual-number structure ($D = y + \varepsilon v$, $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$, combinator ∂).*

Proof. By the converse, every such structure has a solid infinitesimal part; by Theorem 8 the finite solid parts are exactly $\{0, y\}$; by the forward direction these are realised by exactly $\{\text{id}, \partial\}$, and by Lemma 6 the part determines the object $D = y \oplus M$ up to isomorphism. Hence exactly two structures. $\square$

We state Theorem 11 as explicitly conditional, and we do not claim it unconditionally: the unconditional half is Theorem 8, the classification of solid parts, which stands on its own.

7 Comparison with Lanfranchi–Lemay, and significance

Lanfranchi–Lemay (affine schemes)	This note (polynomial functors)
base a commutative ring R ; rigidity needs R a PID	base the rig $\mathbb{N}$ (no subtraction); rigidity needs no PID
projective $\Rightarrow$ free = Rotman structure theorem (uses PID)	linear functor <i>is</i> free $\mathbb{N}$ -module (Lemma 6) — automatic
rank in R -modules, $R^{n^2} \cong R^n$	cardinality rank, $(S \times S) \cong S$ (Lemmas 4, 6)
solid $M \otimes_R M \cong M \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$	solid $M \triangleleft M \cong M \Rightarrow S ^2 = S \Rightarrow S \in \{0, 1\}$
two structures: trivial + Kähler/dual-number extras at non-PID / non-f.g. (Ex. 4.20–4.21)	two parts: $0 + y$, realising $\text{id} + \partial$ extra at infinite support $\mathbb{N} \cdot y$ (§5)

Table 1: The port is faithful (same rank-idempotent mechanism) and sharper on one axis: freeness is automatic over $\mathbb{N}$, so the rigidity lives over a rig rather than a PID. The tensor coincidence (Lemma 4) is the linchpin specific to polynomials.

The classification is a clean deliverable for two audiences. For the tangent-category community [5, 7] it exhibits a rig-based model in which the representable-tangent rigidity holds for a strictly weaker reason than over a PID, and isolates the single structural step (Problem 10) that a rig-internal uniqueness theorem still needs. For the Poly and containers community [1, 2] it upgrades “the container derivative is a tangent structure” [8] toward “it is the *only* nontrivial one,” quantifying exactly how much of that upgrade is proved (the solid-part classification) and how much is reduced to a named lemma (the vertical-lift converse). The shape of the target — *we classified*

the tangent structures on finite-support polynomial functors — is the clean statement; this note delivers its rigidity half.

8 Conclusion

We have proved that a finite-support solid infinitesimal part of $(\mathbf{Poly}, \triangleleft, y)$ has rank in $\{0, 1\}$, hence is either 0 or y , realising the trivial structure and the AAGM container derivative; the arithmetic is Lean-certified with no classical axioms. The one technical observation that makes the port of Lanfranchi–Lemay’s affine-scheme rigidity faithful is that on the linear layer the bundle tensor $\triangleleft$ and the module tensor coincide (Lemma 4), so solidity is measured in the tensor the bundle already uses; and the port is cleaner than its template because over the rig $\mathbb{N}$ freeness is automatic, with no principal-ideal hypothesis. The finiteness is sharp: $\mathbb{N} \cdot y$ is a third solid part, so unqualified uniqueness over $\mathbf{Poly}$ fails.

What remains is a single, precisely located step. Full uniqueness (Theorem 11) follows once one knows that every representable first-order tangent structure has a solid infinitesimal part — the rig-internal analogue of Lanfranchi–Lemay’s vertical-lift universality (Problem 10). We have given two pieces of evidence that it holds: the non-solid finite candidates are exactly the Whitney sums, which are auxiliary rather than primitive; and the derivation that needs the converse is module-theoretic, not ring-theoretic, so the absence of subtraction in $\mathbb{N}$ should not obstruct it. Three routes to close the gap suggest themselves: a direct $\mathbf{Poly}$ -internal proof of the vertical-lift universality; the operadic machinery of Rosický tangent categories of algebras over an operad, the species-adjacent setting closest to polynomial functors; and making “finite support $\iff$ coexponentiable in $(\mathbf{Poly}, \triangleleft)$ ” precise, so the finiteness hypothesis of Theorem 8 becomes intrinsic rather than imposed. Higher Weil algebras $\mathbb{N}[\varepsilon]/\varepsilon^k$ ($k \geq 3$) and iterated tangents are first-order out of scope, as they are for the derivative, for Lanfranchi–Lemay, and for the fibrational theory [4].

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